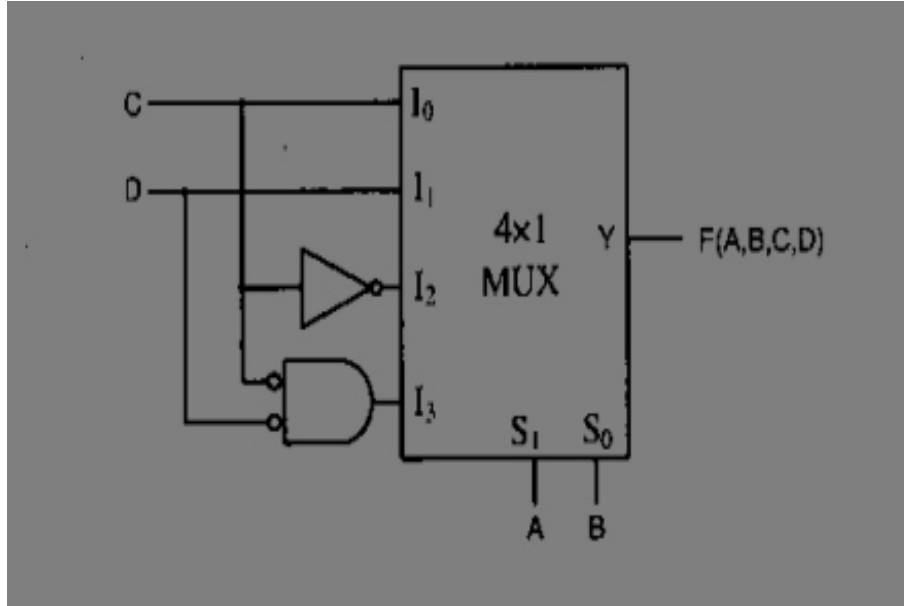


GATE 2010 EC, 39th Question Analysis

Question 39

The Boolean function realized by the logic circuit shown is



A $F = \sum m(0, 1, 3, 5, 9, 10, 14)$

C $F = \sum m(1, 2, 4, 5, 11, 14, 15)$

B $F = \sum m(2, 3, 5, 7, 8, 9, 12, 13)$

D $F = \sum m(2, 3, 5, 7, 8, 9, 12)$

Question Analysis:

- The standard mux equation of 4x1 mux is
- $F = \bar{S}_1 \cdot \bar{S}_0 \cdot I_0 + \bar{S}_1 \cdot S_0 \cdot I_1 + S_1 \cdot \bar{S}_0 \cdot I_2 + S_1 \cdot S_0 \cdot I_3$
- here $S_1 = A, S_0 = B, I_0 = C, I_1 = D, I_2 = \bar{C}, I_3 = \bar{C} \cdot \bar{D}$.
- So $F = \bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot D + A \cdot \bar{B} \cdot \bar{C} + A \cdot B \cdot \bar{C} \cdot \bar{D}$
- The canonical form of F is as follows
- $F = \bar{A} \cdot \bar{B} \cdot C \cdot \bar{D} + \bar{A} \cdot \bar{B} \cdot C \cdot D + \bar{A} \cdot B \cdot \bar{C} \cdot D + \bar{A} \cdot B \cdot C \cdot D + A \cdot \bar{B} \cdot \bar{C} \cdot \bar{D} + A \cdot \bar{B} \cdot \bar{C} \cdot D + A \cdot B \cdot \bar{C} \cdot \bar{D}$
- The Truth table for above expressions is as follows
- From Truth Table we can see that Option-D is correct.

The Truth Table

INPUTS				Output
A	B	C	D	$F(A, B, C, D)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here we used the 7447 and 7474 and displayed the corresponding F value on LED and respective min term on seven segment as per truth table using FSM and verified the expression.

Required Components & Pin Connections

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	7447 IC (1)
4	Seven segment (1)
5	Resistors: 220Ω (1)
6	Jumper Wires
7	USB Cable
8	7474 IC (2)
9	LED (1)

Component	Arduino Pin
Input A(GND or 5v)	Digital 5
Input B(GND or 5v)	Digital 4
Input C(GND or 5v)	Digital 3
Input D(GND or 5v)	Digital 2
Output F (LED)	Digital 8
Output a (7474-1 D1)	Digital 9
Output b (7474-1 D2)	Digital 10
Output c (7474-2 D1)	Digital 11
Output d (7474-2 D2)	Digital 12
Output clk (7474 clk)	Digital 13
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $A = 0, B = 0, C = 0, D = 0$ by connecting them to ground
- $F = \bar{A}.\bar{B}.C + \bar{A}.B.D + A.\bar{B}.\bar{C} + A.B.\bar{C}.\bar{D}$

- No Change inputs A, B, C, D as per truth table by connecting to GND or 5V.
- Now Note Down the Experimental Truth Table

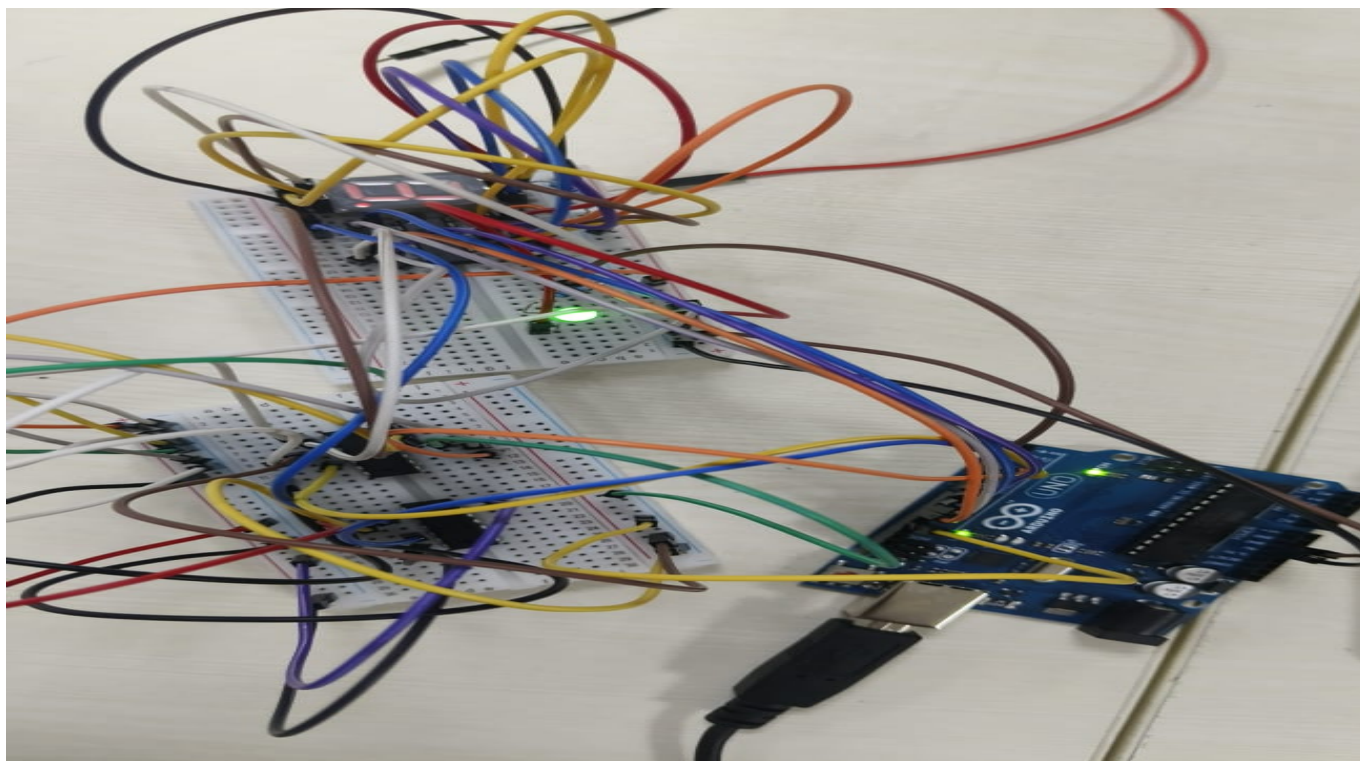
Code Uploading Steps

1. Create a fsm folder
2. Write The code in fsm.asm
3. Run the command "avra fsm.asm". It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precompiled" option
7. Observe the ouput and verify the expression

Experimental Truth Table

INPUTS				Output
A	B	C	D	$F(A, B, C, D)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Circuit



Conclusion

- By comparing Experimental Truth Table and Actual Truth table $F = \sum m(2, 3, 5, 7, 8, 9, 12)$.
- This matches option (D) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.